

FinRelief AI – AI Powered Debt Relief & Financial Recovery Platform

Complete Final Project Report

Cover Page

Project Title: FinRelief AI – AI Powered Debt Relief & Financial Recovery Platform.

Abstract

FinRelief AI is a full-stack web application that assists borrowers with debt management, financial health analysis, settlement prediction, AI-assisted negotiation email generation, and borrower rights awareness using React, FastAPI, SQLite, and Google Gemini AI.

Technology Stack

Frontend: React.js + Vite. Backend: FastAPI (Python). Database: SQLite. API: Axios. Authentication: JWT. Version Control: Git & GitHub.

Brainstorming & Ideation

Identified the need for an AI-assisted debt recovery platform and defined objectives, scope, and expected outcomes.

Requirement Analysis

Specified functional and non-functional requirements, software and hardware requirements, and user expectations.

Project Design

Designed the system architecture, database schema, REST APIs, security mechanisms, and responsive user interface.

Project Planning

Applied Agile methodology using epics, user stories, sprint planning, milestones, and risk management.

Project Development

Implemented authentication, dashboard, loan management, financial health, settlement prediction, negotiation email, borrower rights, history, AI integration, and GitHub version control.

Project Testing

Performed unit, integration, system, error handling, and performance testing to validate application reliability.

Project Documentation

Prepared README, installation guide, API documentation, database documentation, GitHub repository documentation, and maintenance guidelines.

Project Demonstration

Demonstrated complete application workflow from login to AI-powered financial assistance and deployment readiness.

Results

The application successfully achieved its objectives by providing a secure, responsive, modular, and AI-assisted financial recovery platform.

Future Scope

Potential enhancements include MySQL, cloud deployment, advanced AI recommendations, PDF reports, email notifications, and mobile application support.

Conclusion

FinRelief AI demonstrates modern full-stack development practices and provides a scalable foundation for intelligent debt management solutions.

References

React.js, Vite, FastAPI, Python, SQLite, Axios, Git, GitHub, Google Gemini API, official documentation.